

SPESIFIKASI STRUKTUR DATABASE FISIK

KOPERASI SULFINDO

Status Dokumen:	Production-Ready / As-Built
Target Database:	MySQL (Prisma ORM engine)
Dibuat Oleh:	Senior Software Engineer & Technical Architect
Tanggal Rilis:	22 Mei 2026

RANCANGAN STRUKTUR DATABASE - KOPERASI SULFINDO

Senior Database Engineer & Technical Architect Perspective

Dokumen ini menyajikan spesifikasi arsitektur database Koperasi Sulfindo berbasis database MySQL (dengan Prisma ORM). Desain ini dioptimalkan untuk performa tinggi, auditabilitas transaksi keuangan (*double-entry ledger*), dan integritas data yang kokoh.

1. PETA MODUL & DAFTAR TABEL

Struktur database dibagi menjadi 7 domain utama untuk memisahkan tanggung jawab data (*Separation of Concerns*):

- 1. **Core & Membership:** `users`, `sessions`, `members`, `units`, `audit_logs`
- 2. **Double-Entry Ledger (Akuntansi):** `chart_of_accounts`, `journal_entries`, `journal_lines`, `monthly_closures`
- 3. **Simpan Pinjam (Savings & Loans):** `saving_types`, `savings`, `saving_transactions`, `loan_products`, `loan_applications`, `loans`, `loan_schedules`, `loan_payments`
- 4. **Point of Sale (POS) & Inventaris:** `products`, `product_categories`, `orders`, `order_items`, `order_payments`, `order_returns`, `warehouse_locations`, `stock_balances`, `stock_reorder_points`, `stock_movements`, `product_costing`
- 5. **Manajemen Stok & Distribusi:** `stock_transfer_orders`, `stock_transfer_items`, `stock_opname`, `stock_opname_details`
- 6. **Konsinyasi & Procurement:** `suppliers`, `consignment_items`, `consignment_payables`, `consignment_settlements`, `purchase_orders`, `purchase_order_items`, `good_receipts`, `good_receipt_items`
- 7. **Kuangan Tambahan, RAT & SHU:** `accounts_payable`, `accounts_payable_details`, `accounts_receivable`, `accounts_receivable_details`, `tax_calculations`, `shu_periods`, `shu_distributions`, `rat_attendances`, `loyalty_programs`, `loyalty_memberships`, `loyalty_redemptions`, `price_tiers`, `customer_segments`, `sales_analytics`, `product_analytics`, `budgets`, `promotions`

2. DETAIL TABEL RELASI & KAMUS DATA

MODUL 1: CORE & MEMBERSHIP

A. Tabel `units` (Unit Kerja/Usaha)

Menampung unit kerja anggota atau unit usaha bisnis koperasi (Simpan Pinjam, Toko, PPOB, dll.).

- **Primary Key:** `id` (BigInt Unsigned Auto-Increment)
- **Foreign Key:** `parent_id` -> `units.id` (Relasi self-referencing 1:N, Nullable, untuk struktur hierarki unit)
- **Kamus Data & Constraints:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
<code>id</code>	BigInt Unsigned	PK, Not Null, Unique		ID internal unit
<code>code</code>	VarChar(20)	Unique, Not Null		Kode unik unit (misal: USP, UTK)
<code>name</code>	VarChar(150)	Not Null		Nama unit usaha

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
type	Enum	Not Null	simpan_pinjam	simpan_pinjam, toko, ppob, elearning, asuransi, investasi, induk
parent_id	BigInt Unsigned	Nullable		ID unit induk jika sub-unit
address	VarChar(255)	Nullable		Alamat fisik unit
phone	VarChar(20)	Nullable		Nomor telepon unit
is_active	Boolean	Not Null	true	Status keaktifan operasional
created_at	Timestamp	Nullable		Waktu record dibuat
updated_at	Timestamp	Nullable		Waktu record diperbarui

- Relasi:

- **units** (Parent) 1 : N **units** (Sub-unit)

- **units** 1 : N **members**

- **units** 1 : N **chart_of_accounts**

- **units** 1 : N **products**

B. Tabel **members** (Profil Anggota)

Menyimpan profil personal dan kepegawaian seluruh anggota koperasi Sulfindo.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)

- **Foreign Key:** **unit_id** -> **units.id** (Relasi 1:N, Not Null)

- **Kamus Data & Constraints:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null, Unique		ID internal anggota
member_code	VarChar(20)	Unique, Not Null		Nomor Anggota Unik Koperasi
nik	VarChar(16)	Unique, Not Null		Nomor Induk Kependudukan
full_name	VarChar(150)	Not Null		Nama Lengkap sesuai KTP
email	VarChar(100)	Unique, Nullable		Email pribadi
phone	VarChar(20)	Nullable		Nomor telepon / HP
address	VarChar(255)	Nullable		Alamat tinggal
birth_date	Date	Nullable		Tanggal lahir
gender	Enum	Nullable		male, female
join_date	Date	Not Null		Tanggal bergabung
photo_path	VarChar(255)	Nullable		Path file foto profil
status	Enum	Not Null	active	active, inactive, suspended
unit_id	BigInt Unsigned	FK, Not Null		Referensi unit kerja perusahaan
created_at	Timestamp	Nullable		Waktu dibuat
updated_at	Timestamp	Nullable		Waktu diupdate
deleted_at	Timestamp	Nullable		Soft delete timestamp

- Relasi & Kardinalitas:

- **units** 1 : N **members** (1 unit kerja diisi banyak anggota)

- **members** 1 : 1 **users** (Anggota memiliki maksimal 1 akun login)

- **members** 1 : N **savings** (Anggota memiliki banyak akun tabungan)
- **members** 1 : N **loans** (Anggota memiliki banyak akun pinjaman)

C. Tabel **users** (Akun Autentikasi)

Menangani kredensial sistem dan otorisasi menu.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)
- **Foreign Key:** **member_id** -> **members.id** (Relasi 1:1, Nullable, Unique)
- **Kamus Data & Constraints:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal user
member_id	BigInt Unsigned	FK, Unique, Nullable		Link ke profil anggota
username	VarChar(50)	Unique, Not Null		Username login
email	VarChar(100)	Unique, Not Null		Email login
password	VarChar(255)	Not Null		Password (hash bcrypt)
role	Enum	Not Null	anggota	superadmin, admin, pengurus, kasir, anggota
is_active	Boolean	Not Null	true	Status aktif akun login
mfa_secret	VarChar(100)	Nullable		Kunci Google Authenticator TOTP
mfa_enabled	Boolean	Not Null	false	Status status aktif MFA
last_login_at	Timestamp	Nullable		Waktu login terakhir
last_login_ip	VarChar(45)	Nullable		IP login terakhir
remember_token	VarChar(100)	Nullable		Cookie remember me token

- **Relasi & Kardinalitas:**
- **members** 1 : 1 **users**
- **users** 1 : N **audit_logs**
- **users** 1 : N **journal_entries** (User sebagai petugas poster jurnal)
- **users** 1 : N **saving_transactions** (User sebagai kasir pemroses simpanan)
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MODUL 2: BUKU BESAR AKUNTANSI (DOUBLE-ENTRY LEDGER)

A. Tabel **chart_of_accounts** (Bagan Akun / COA)

Daftar rekening akuntansi standar koperasi.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)
- **Foreign Key:**
- **unit_id** -> **units.id** (Relasi 1:N, Not Null)
- **parent_id** -> **chart_of_accounts.id** (Relasi self 1:N, Nullable, untuk sub-account)
- **Kamus Data & Constraints:**
- **Constraint Unik:** Kombinasi [**unit_id**, **code**] wajib unik (**uq_unit_account_code**), mencegah kode akun ganda dalam satu unit bisnis.

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal akun

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
unit_id	BigInt Unsigned	FK, Not Null		Lingkup unit usaha akun
code	VarChar(20)	Not Null		Kode nomor COA (misal: 11110)
name	VarChar(150)	Not Null		Nama akun (misal: Kas Utama)
type	Enum	Not Null		asset, liability, equity, revenue, expense
normal_balance	Enum	Not Null		debit, credit
parent_id	BigInt Unsigned	FK, Nullable		Akun header tingkat atas
level	SmallInt	Not Null	1	Kedalaman struktur akun (1-5)
is_header	Boolean	Not Null	false	True jika akun penampung (bukan transaksi)
is_active	Boolean	Not Null	true	Status akun aktif digunakan

- Relasi:

- **units** 1 : N **chart_of_accounts**

- **chart_of_accounts** (Header) 1 : N **chart_of_accounts** (Detail)

- **chart_of_accounts** 1 : N **journal_lines**

B. Tabel **journal_entries** (Header Jurnal Keuangan)

Kepala jurnal transaksi untuk semua transaksi finansial koperasi.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)

- **Foreign Key:**

- **unit_id** -> **units.id** (Relasi 1:N, Not Null)

- **posted_by** -> **users.id** (Relasi 1:N, Nullable)

- **reversed_by_entry_id** -> **journal_entries.id** (Relasi 1:1, Nullable, untuk jurnal balik/koreksi)

- **Kamus Data & Constraints:**

- **Constraint Unik:** **entry_no** wajib unik global (Kode Jurnal otomatis).

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal jurnal
unit_id	BigInt Unsigned	FK, Not Null		Unit asal transaksi keuangan
entry_no	VarChar(60)	Unique, Not Null		Nomor Bukti Jurnal Unik (misal: JR/USP/2026/05/001)
entry_date	Date	Not Null		Tanggal pencatatan akuntansi
description	VarChar(255)	Not Null		Keterangan/narasi transaksi
reference	VarChar(100)	Nullable		Referensi eksternal (Nomor Struk POS/Pinjaman)
source	Enum	Not Null	manual	manual, saving, loan, loan_payment, pos, ppob, shu, adjustment
posted_by	BigInt Unsigned	FK, Nullable		Pembuat/penyetuju jurnal
posted_at	Timestamp	Nullable		Waktu posting
is_posted	Boolean	Not Null	false	True jika sudah dikunci (masuk buku besar)
is_reversed	Boolean	Not Null	false	True jika jurnal telah dibatalkan/dibalik
reversed_by_entry_id	BigInt Unsigned	FK, Nullable		Link ke jurnal pembatalnya

C. Tabel `journal_lines` (Baris Jurnal / Debit-Kredit)

Baris rincian debit-kredit (*double-entry*) dari suatu jurnal.

- **Primary Key:** `id` (BigInt Unsigned Auto-Increment)
- **Foreign Key:**
 - `journal_id` -> `journal_entries.id` (Relasi 1:N dengan **ON DELETE CASCADE**, Not Null)
 - `account_id` -> `chart_of_accounts.id` (Relasi 1:N, Not Null)
- **Kamus Data & Constraints:**
- **Aturan Integritas:** Jumlah nominal `debit` dan `credit` dalam satu `journal_id` wajib bernilai seimbang (diperiksa pada level application service).

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
<code>id</code>	BigInt Unsigned	PK, Not Null		ID internal baris
<code>journal_id</code>	BigInt Unsigned	FK, Not Null		Link ke header jurnal
<code>account_id</code>	BigInt Unsigned	FK, Not Null		Akun COA yang dipengaruhi
<code>debit</code>	Decimal(15,2)	Not Null	0.00	Nilai debit transaksi
<code>credit</code>	Decimal(15,2)	Not Null	0.00	Nilai kredit transaksi
<code>description</code>	VarChar(255)	Nullable		Memo rincian per baris akun

MODUL 3: SIKLUS SIMPAN PINJAM (SAVINGS & LOANS)**A. Tabel `savings` (Rekening Simpanan Anggota)**

Menyimpan saldo agregat simpanan per anggota per jenis simpanan.

- **Primary Key:** `id` (BigInt Unsigned Auto-Increment)
- **Foreign Key:**
 - `member_id` -> `members.id` (Relasi 1:N, Not Null)
 - `saving_type_id` -> `saving_types.id` (Relasi 1:N, Not Null)
- **Constraint Unik:** [`member_id`, `saving_type_id`] wajib unik (**uq_member_saving_type**) agar tidak terjadi rekening ganda untuk jenis tabungan yang sama pada satu anggota.
- **Kamus Data:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
<code>id</code>	BigInt Unsigned	PK, Not Null		ID internal rekening
<code>member_id</code>	BigInt Unsigned	FK, Not Null		Pemilik tabungan
<code>saving_type_id</code>	BigInt Unsigned	FK, Not Null		Jenis Simpanan (Pokok/Wajib/Sukarela)
<code>balance</code>	Decimal(15,2)	Not Null	0.00	Saldo saat ini (Rupiah)
<code>total_deposit</code>	Decimal(15,2)	Not Null	0.00	Akumulasi setoran historis
<code>total_withdraw</code>	Decimal(15,2)	Not Null	0.00	Akumulasi penarikan historis

B. Tabel `saving_transactions` (Mutasi Tabungan)

Jurnal mutasi kas simpanan anggota untuk kebutuhan transparansi audit (*audit trail*).

- **Primary Key:** `id` (BigInt Unsigned Auto-Increment)
- **Foreign Key:**

- **savings_id** -> **savings.id** (Relasi 1:N, Not Null)
- **member_id** -> **members.id** (Relasi 1:N, Not Null)
- **processed_by** -> **users.id** (Relasi 1:N, Nullable)
- **Kamus Data & Constraints:**
- **Constraint Unik:** **reference_no** unik untuk melacak dokumen transaksi (misal: TXS/2026/05/0001).

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID mutasi
savings_id	BigInt Unsigned	FK, Not Null		Akun simpanan tujuan
member_id	BigInt Unsigned	FK, Not Null		Anggota pemilik
type	Enum	Not Null		deposit, withdraw, salary_cut, shu_credit, loan_deduct, pos_deduct, adjustment
amount	Decimal(15,2)	Not Null		Nominal transaksi
balance_before	Decimal(15,2)	Not Null		Saldo sebelum transaksi
balance_after	Decimal(15,2)	Not Null		Saldo sesudah transaksi
reference_no	VarChar(60)	Unique, Not Null		Nomor transaksi mutasi unik
note	Text	Nullable		Catatan penjelas
processed_by	BigInt Unsigned	FK, Nullable		Petugas pemroses transaksi
transaction_at	Timestamp	Not Null		Waktu transaksi terjadi

C. Tabel **loans** (Akun Kontrak Pinjaman)

Kontrak pinjaman legal yang disetujui untuk anggota.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)
- **Foreign Key:**
- **application_id** -> **loan_applications.id** (Relasi 1:1, Not Null, Unique)
- **member_id** -> **members.id** (Relasi 1:N, Not Null)
- **Kamus Data & Constraints:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal pinjaman
application_id	BigInt Unsigned	FK, Unique, Not Null		Link ke formulir pengajuan
member_id	BigInt Unsigned	FK, Not Null		Anggota peminjam
loan_no	VarChar(60)	Unique, Not Null		Nomor akad pinjaman unik
principal	Decimal(15,2)	Not Null		Nominal pokok pinjaman (Plafon)
interest_rate	Decimal(5,2)	Not Null		Suku bunga per periode (persen)
interest_method	Enum	Not Null		flat, anuitas, efektif
admin_fee	Decimal(15,2)	Not Null	0.00	Potongan biaya administrasi awal
tenor_months	SmallInt	Not Null		Jangka waktu angsuran (bulan)
disbursed_at	Date	Not Null		Tanggal pencairan dana
first_due_date	Date	Not Null		Tanggal cicilan pertama
last_due_date	Date	Not Null		Tanggal cicilan berakhir

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
monthly_installment	Decimal(15,2)	Not Null		Kewajiban angsuran total per bulan
outstanding_principal	Decimal(15,2)	Not Null		Sisa piutang pokok berjalan
total_paid	Decimal(15,2)	Not Null	0.00	Total dana yang sudah dibayarkan
repayment_method	Enum	Not Null	cash	cash, salary_cut, saving_deduct
status	Enum	Not Null	active	active, paid_off, overdue, restructured, written_off

D. Tabel **loan_schedules** (Jadwal Angsuran Pinjaman)

Rencana pembayaran amortisasi bulanan pinjaman.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)
- **Foreign Key:** **loan_id** -> **loans.id** (Relasi 1:N dengan **ON DELETE CASCADE**, Not Null)
- **Constraints:** Kombinasi [**loan_id, installment_no**] wajib unik (**uq_loan_installment**).
- **Kamus Data:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal jadwal
loan_id	BigInt Unsigned	FK, Not Null		Link ke kontrak pinjaman
installment_no	SmallInt	Not Null		Nomor angsuran ke-N (misal: 1 s/d 12)
due_date	Date	Not Null		Tanggal jatuh tempo
principal_due	Decimal(15,2)	Not Null		Angsuran pokok yang wajib dibayar
interest_due	Decimal(15,2)	Not Null		Jasa bunga yang wajib dibayar
total_due	Decimal(15,2)	Not Null		Total kewajiban (principal_due + interest_due)
principal_paid	Decimal(15,2)	Not Null	0.00	Pokok yang sudah dibayar secara riil
interest_paid	Decimal(15,2)	Not Null	0.00	Bunga yang sudah dibayar secara riil
penalty_paid	Decimal(15,2)	Not Null	0.00	Denda terlambat yang sudah dibayar
paid_at	Timestamp	Nullable		Tanggal pelunasan angsuran ini
status	Enum	Not Null	pending	pending, partial, paid, overdue

- **Korelasi Agregasi PHU & Laba Rugi:**

Bunga Jasa (**interest_paid**) dan Denda (**penalty_paid**) dari tabel ini ditarik sebagai pendapatan operasional berjalan koperasi.

MODUL 4: POINT OF SALE (POS) & INVENTARIS TOKO

A. Tabel **products** (Katalog Produk Waserda)

Master data barang dagangan Toko Waserda Koperasi Sulfindo.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)
- **Foreign Key:**
- **unit_id** -> **units.id** (Relasi 1:N, Not Null)
- **category_id** -> **product_categories.id** (Relasi 1:N, Not Null)

- Kamus Data & Constraints:

- **Constraint Unik:** **sku** (Barcode / Stock Keeping Unit) wajib unik.

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal produk
unit_id	BigInt Unsigned	FK, Not Null		Unit toko kepemilikan barang
category_id	BigInt Unsigned	FK, Not Null		Kategori produk
sku	VarChar(50)	Unique, Not Null		Barcode/SKU barang
name	VarChar(200)	Not Null		Nama produk (misal: Minyak Goreng 2L)
description	Text	Nullable		Deskripsi produk
purchase_price	Decimal(15,2)	Not Null	0.00	Harga beli modal dari supplier
price	Decimal(15,2)	Not Null		Harga jual umum retail
member_price	Decimal(15,2)	Nullable		Harga jual khusus anggota koperasi
stock	Int	Not Null	0	Stok global (virtual sync)
min_stock	Int	Not Null	0	Batas stok minimum peringatan restock
unit_measure	VarChar(20)	Not Null	pcs	Satuan barang (pcs, pack, kg)
is_active	Boolean	Not Null	true	Status aktif dijual
is_online	Boolean	Not Null	true	Status tayang di e-commerce anggota
restock_requested	Boolean	Not Null	false	Penanda butuh order ulang

B. Tabel **stock_balances (Saldo Stok Fisik Per Gudang/Lokasi)**

Pencatatan inventaris multi-lokasi (*multi-warehouse inventory*).

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)

- **Foreign Key:**

- **product_id** -> **products.id** (Relasi 1:N dengan **ON DELETE CASCADE**, Not Null)

- **location_id** -> **warehouse_locations.id** (Relasi 1:N dengan **ON DELETE CASCADE**, Not Null)

- **Constraint Unik:** Kombinasi [**product_id**, **location_id**] unik.

- **Kamus Data:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal saldo
product_id	BigInt Unsigned	FK, Not Null		ID produk
location_id	BigInt Unsigned	FK, Not Null		ID gudang/lokasi retail
qty_on_hand	Int	Not Null	0	Stok fisik total di lokasi tersebut
qty_reserved	Int	Not Null	0	Stok dipesan (tertahan di checkout POS/online)
qty_available	Int	Not Null	0	Stok riil siap jual (qty_on_hand - qty_reserved)

C. Tabel **stock_movements (Mutasi / Kartu Stok)**

Log terperinci keluar masuknya barang dagangan untuk audit stock mutasi.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)

- **Foreign Key:**

- **product_id** -> **products.id** (Relasi 1:N, Not Null)

- **created_by** -> **users.id** (Relasi 1:N, Nullable)

- **Kamus Data:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal mutasi stok
product_id	BigInt Unsigned	FK, Not Null		ID produk yang bermutasi
type	Enum	Not Null		in, out, adjustment, return, transfer
qty	Int	Not Null		Kuantitas mutasi (bisa positif/negatif)
stock_before	Int	Not Null		Saldo stok global sebelum mutasi
stock_after	Int	Not Null		Saldo stok global sesudah mutasi
reference	VarChar(100)	Nullable		Dokumen referensi (Order No / Opname No)
note	Text	Nullable		Catatan tambahan
created_by	BigInt Unsigned	FK, Nullable		User penanggung jawab mutasi

MODUL 5: KONSINYASI (TITIP JUAL BARANG SUPPLIER)

Sistem konsinyasi di Koperasi Sulfindo melacak barang titip jual milik pihak ketiga (supplier).

A. Tabel **consignment_items** (Penerimaan Barang Konsinyasi)

Mencatat penerimaan fisik barang konsinyasi.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)

- **Foreign Key:**

- **product_id** -> **products.id** (Relasi 1:N, Not Null)

- **supplier_id** -> **suppliers.id** (Relasi 1:N, Not Null)

- **Kamus Data:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID penerimaan konsinyasi
product_id	BigInt Unsigned	FK, Not Null		ID barang titip
supplier_id	BigInt Unsigned	FK, Not Null		Supplier pemilik barang
consignment_date	Date	Not Null		Tanggal penitipan barang
qty_received	Int	Not Null		Jumlah barang yang dititipkan
qty_sold	Int	Not Null	0	Jumlah barang yang sudah laku terjual
qty_returned	Int	Not Null	0	Jumlah barang sisa yang diretur ke supplier
return_reason	VarChar(500)	Nullable		Alasan pengembalian barang
return_date	Timestamp	Nullable		Tanggal barang diretur
status	Enum	Not Null	active	active, returned, settled, closed

B. Tabel **consignment_payables** (Hutang Konsinyasi Terjual)

Menampung kewajiban bayar koperasi atas barang konsinyasi yang sudah laku terjual dan belum dibayar.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)

- **Foreign Key:**

- **consignment_id** -> **consignment_items.id** (Relasi 1:N dengan **ON DELETE CASCADE**, Not Null)
- **supplier_id** -> **suppliers.id** (Relasi 1:N, Not Null)
- Kamus Data:

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID hutang konsinyasi
consignment_id	BigInt Unsigned	FK, Not Null		Link ke penerimaan item
supplier_id	BigInt Unsigned	FK, Not Null		Supplier yang berhak menerima uang
qty_sold	Int	Not Null		Kuantitas barang yang terjual
unit_price	Decimal(15,2)	Not Null		Harga kesepakatan bagi hasil per unit
total_amount	Decimal(15,2)	Not Null		Total hutang (qty_sold * unit_price)
status	Enum	Not Null	pending	pending, partially_paid, paid

C. Tabel **consignment_settlements** (Pelunasan Pembayaran Konsinyasi)

Pencatatan realisasi pembayaran bagi hasil ke supplier.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)
- **Foreign Key:**
- **payable_id** -> **consignment_payables.id** (Relasi 1:N dengan **ON DELETE CASCADE**, Not Null)
- **processed_by** -> **users.id** (Relasi 1:N, Not Null)
- Kamus Data:

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID pelunasan
payable_id	BigInt Unsigned	FK, Not Null		Referensi hutang yang dibayar
settlement_no	VarChar(60)	Unique, Not Null		Nomor bukti pembayaran konsinyasi
settlement_date	Date	Not Null		Tanggal pembayaran
amount_paid	Decimal(15,2)	Not Null		Jumlah dana ditransfer
payment_method	Enum	Not Null		cash, check, transfer, debit_card, credit_card, qris
reference_no	VarChar(100)	Nullable		Nomor referensi bank/bukti transfer
processed_by	BigInt Unsigned	FK, Not Null		Petugas keuangan pembayar

MODUL 6: AP & AR (HUTANG/PIUTANG USAHA & PROCUREMENT)

A. Tabel **accounts_payable** (Hutang Dagang / AP)

Mencatat tagihan masuk dari supplier atas pembelian stok retail non-konsinyasi.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)
- **Foreign Key:** **supplier_id** -> **suppliers.id** (Relasi 1:N, Not Null)
- Kamus Data:

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID tagihan
supplier_id	BigInt Unsigned	FK, Not Null		Supplier penagih

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
invoice_no	VarChar(60)	Unique, Not Null		Nomor invoice dari supplier
invoice_date	Date	Not Null		Tanggal diterbitkan
due_date	Date	Not Null		Tanggal jatuh tempo
subtotal	Decimal(15,2)	Not Null		Total bersih barang
tax_amount	Decimal(15,2)	Not Null	0.00	PPN/pajak pembelian
total_amount	Decimal(15,2)	Not Null		Total tagihan kotor
amount_paid	Decimal(15,2)	Not Null	0.00	Total yang sudah diangsur/dibayar
amount_due	Decimal(15,2)	Not Null		Sisa hutang dagang
status	Enum	Not Null	open	open, partial, paid, overdue, cancelled

B. Tabel **accounts_receivable** (Piutang Non-Pinjaman / AR)

Mencatat piutang toko (seperti transaksi Paylater anggota / instansi luar).

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)
- **Foreign Key:** **member_id** -> **members.id** (Relasi 1:N, Nullable)
- **Kamus Data:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID piutang
member_id	BigInt Unsigned	FK, Nullable		Link ke anggota jika diutangkan ke anggota
customer_name	VarChar(150)	Not Null		Nama pembeli/pihak ketiga
invoice_no	VarChar(60)	Unique, Not Null		Nomor invoice piutang
invoice_date	Date	Not Null		Tanggal tagihan
due_date	Date	Not Null		Tanggal jatuh tempo
total_amount	Decimal(15,2)	Not Null		Total piutang dagang
amount_paid	Decimal(15,2)	Not Null	0.00	Total cicilan piutang diterima
amount_due	Decimal(15,2)	Not Null		Sisa piutang dagang
status	Enum	Not Null	open	open, partial, paid, overdue, cancelled

MODUL 7: RAT & SHU (DISTRIBUSI LABA TAHUNAN)

A. Tabel **shu_periods** (Periode Buku RAT & Alokasi Makro)

Menyimpan data rekapitulasi laba bersih tahunan dan konfigurasi persentase pembagian SHU.

- **Primary Key:** **id** (BigInt Unsigned Auto-Increment)
- **Foreign Key:** **unit_id** -> **units.id** (Relasi 1:N, Not Null)
- **Constraint Unik:** [**unit_id, period_year**] wajib unik (**uq_unit_shu_year**).
- **Kamus Data:**

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal periode SHU

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
unit_id	BigInt Unsigned	FK, Not Null		Lingkup unit usaha penanggung
period_year	Year (Int)	Not Null		Tahun Buku RAT (misal: 2026)
total_revenue	Decimal(15,2)	Not Null	0.00	Total Pendapatan Operasional
total_expense	Decimal(15,2)	Not Null	0.00	Total Beban Operasional
total_shu	Decimal(15,2)	Not Null	0.00	SHU Bersih Koperasi (Rev - Exp)
shu_for_member	Decimal(15,2)	Not Null	0.00	Porsi untuk pembagian anggota (umumnya 55%)
shu_for_reserve	Decimal(15,2)	Not Null	0.00	Porsi dana cadangan koperasi (20%)
shu_for_pengurus	Decimal(15,2)	Not Null	0.00	Porsi insentif Pengurus & Pengawas (5%)
shu_for_education	Decimal(15,2)	Not Null	0.00	Porsi dana pendidikan (5%)
status	Enum	Not Null	draft	draft, calculated, approved, distributed, closed

B. Tabel shu_distributions (Detail Pembagian Per Anggota)

Detail nominal SHU riil yang didapatkan oleh masing-masing anggota.

- Primary Key: id (BigInt Unsigned Auto-Increment)
- Foreign Key:
 - shu_period_id -> shu_periods.id (Relasi 1:N, Not Null)
 - member_id -> members.id (Relasi 1:N, Not Null)
- Constraint Unik: [shu_period_id, member_id] wajib unik (uq_shu_member) untuk mencegah pembagian ganda pada anggota yang sama di tahun buku yang sama.
- Kamus Data:

Nama Kolom	Tipe Data	Constraints	Default	Keterangan
id	BigInt Unsigned	PK, Not Null		ID internal detail
shu_period_id	BigInt Unsigned	FK, Not Null		ID tahun buku
member_id	BigInt Unsigned	FK, Not Null		Anggota penerima
savings_weight	Decimal(10,4)	Not Null	0.0000	Persentase kontribusi simpanan anggota
activity_weight	Decimal(10,4)	Not Null	0.0000	Persentase keaktifan belanja/pinjam anggota
shu_amount	Decimal(15,2)	Not Null		Nominal bersih SHU didapat (Rupiah)
status	Enum	Not Null	pending	pending, disbursed
disbursement_method	Enum	Nullable		saving_credit, cash, transfer
disbursed_at	Timestamp	Nullable		Waktu pencairan dana

3. LOGICAL SCHEMA & ALUR KORELASI DATA

Visualisasi hubungan antarentitas utama dalam sistem Koperasi Sulfindo disajikan melalui diagram ERD di bawah ini:

erDiagram

UNITS ||--o{ MEMBERS : "memiliki"

UNITS ||--o{ CHART_OF_ACCOUNTS : "mempunyai"

MEMBERS ||--|| USERS : "memiliki"

MEMBERS ||--o{ SAVINGS : "menyimpan"

MEMBERS ||--o{ LOANS : "meminjam"

MEMBERS ||--o{ ORDERS : "melakukan belanja"

CHART_OF_ACCOUNTS ||--o{ JOURNAL_LINES : "terlibat di"

JOURNAL_ENTRIES ||--|{ JOURNAL_LINES : "berisi"

SAVING_TYPES ||--o{ SAVINGS : "menentukan tipe"

SAVINGS ||--o{ SAVING_TRANSACTIONS : "mencatat mutasi"

LOAN_PRODUCTS ||--o{ LOAN_APPLICATIONS : "berbasis produk"

LOAN_APPLICATIONS ||--|| LOANS : "menghasilkan"

LOANS ||--|{ LOAN_SCHEDULES : "memiliki rencana"

LOANS ||--o{ LOAN_PAYMENTS : "memiliki bukti bayar"

LOAN_SCHEDULES ||--o{ LOAN_PAYMENTS : "dilunasi oleh"

PRODUCTS ||--o{ ORDER_ITEMS : "dijual di"

PRODUCTS ||--o{ STOCK_BALANCES : "memiliki stok di"

PRODUCTS ||--o{ STOCK_MOVEMENTS : "memiliki history"

WAREHOUSE_LOCATIONS ||--o{ STOCK_BALANCES : "menampung"

ORDERS ||--|{ ORDER_ITEMS : "berisi"

ORDERS ||--o{ ORDER_PAYMENTS : "dibayar dengan"

ORDERS ||--o{ ORDER_RETURNS : "dapat diretur"

SUPPLIERS ||--o{ CONSIGNMENT_ITEMS : "menitipkan"

```
CONSIGNMENT_ITEMS ||--|{ CONSIGNMENT_PAYABLES : "menghasilkan hutang"
```

```
CONSIGNMENT_PAYABLES ||--o{ CONSIGNMENT_SETTLEMENTS : "dilunasi"
```

3. LOGICAL & PHYSICAL SCHEMA PER SUB-SISTEM

Untuk menggambarkan sistem Koperasi Sulfindo yang sedang berjalan secara visual dan mendalam, diagram hubungan antarentitas (ERD) di bawah ini dipecah menjadi sub-sistem utama dengan memuat daftar field, tipe data, constraint, serta kardinalitas relasi secara fisik sesuai dengan file **schema.prisma** yang aktif.

A. Core & Membership Sub-System

Diagram ini mencakup struktur unit kerja, data profil anggota, autentikasi user, sesi aktif, serta log audit keamanan sistem.

erDiagram

users {

uint64 id PK "unsigned auto_increment"

uint64 member_id FK "nullable, unique"

varchar_50 username "unique"

varchar_100 email "unique"

varchar_255 password

enum_users_role role "default: anggota"

boolean is_active "default: true"

varchar_100 mfa_secret "nullable"

boolean mfa_enabled "default: false"

timestamp last_login_at "nullable"

varchar_45 last_login_ip "nullable"

varchar_100 remember_token "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

timestamp deleted_at "nullable"

}

members {

uint64 id PK "unsigned auto_increment"

varchar_20 member_code "unique"

varchar_16 nik "unique"

varchar_150 full_name

varchar_100 email "unique, nullable"

varchar_20 phone "nullable"

varchar_255 address "nullable"

date birth_date "nullable"

enum_members_gender gender "nullable"


```

date join_date

varchar_255 photo_path "nullable"

enum_members_status status "default: active"

uint64 unit_id FK "unsigned"

timestamp created_at "nullable"

timestamp updated_at "nullable"

timestamp deleted_at "nullable"

}

units {

uint64 id PK "unsigned auto_increment"

varchar_20 code "unique"

varchar_150 name

enum_units_type type "default: simpan_pinjam"

uint64 parent_id FK "nullable, unsigned"

varchar_255 address "nullable"

varchar_20 phone "nullable"

boolean is_active "default: true"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

sessions {

varchar_255 id PK "cuid"

uint64 user_id FK "nullable, unsigned"

varchar_45 ip_address "nullable"

text user_agent "nullable"

longtext payload

int last_activity

}

```

```

audit_logs {

uint64 id PK "unsigned auto_increment"

uint64 user_id FK "nullable, unsigned"

varchar_50 action

varchar_100 model_type "nullable"

uint64 model_id FK "nullable, unsigned"

json old_values "nullable"

json new_values "nullable"

varchar_45 ip_address "nullable"

varchar_255 user_agent "nullable"

varchar_500 url "nullable"

timestamp created_at

}

units ||--o{ units : "parent_id"

units ||--o{ members : "unit_id"

members ||--o{ users : "member_id"

users ||--o{ audit_logs : "user_id"

users ||--o{ sessions : "user_id"

```

B. Double-Entry Accounting & Ledger Sub-System

Mengatur bagan akun (COA), jurnal transaksi masuk, detail baris jurnal debit/kredit seimbang, serta proses tutup buku bulanan.

```
erDiagram
```

```
chart_of_accounts {
```

```
uint64 id PK "unsigned auto_increment"
```

```
uint64 unit_id FK "unsigned"
```

```
varchar_20 code
```

```
varchar_150 name
```

```
enum_coa_type type
```

```
enum_normal_balance normal_balance
```

```
uint64 parent_id FK "nullable, unsigned"
```

```
int16 level "default: 1"
```

```
boolean is_header "default: false"
```

```
boolean is_active "default: true"
```

```
timestamp created_at "nullable"
```

```
timestamp updated_at "nullable"
```

```
}
```

```
journal_entries {
```

```
uint64 id PK "unsigned auto_increment"
```

```
uint64 unit_id FK "unsigned"
```

```
varchar_60 entry_no "unique"
```

```
date entry_date
```

```
varchar_255 description
```

```
varchar_100 reference "nullable"
```

```
enum_journal_source source "default: manual"
```

```
uint64 posted_by FK "nullable, unsigned"
```

```
timestamp posted_at "nullable"
```

```
boolean is_posted "default: false"
```

```
boolean is_reversed "default: false"
```

```
uint64 reversed_by_entry_id FK "nullable, unsigned"
```

```

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

journal_lines {

uint64 id PK "unsigned auto_increment"

uint64 journal_id FK "unsigned"

uint64 account_id FK "unsigned"

decimal debit "15, 2, default: 0.00"

decimal credit "15, 2, default: 0.00"

varchar_255 description "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

monthly_closures {

uint64 id PK "unsigned auto_increment"

int16 month

int16 year

decimal total_revenue "15, 2, default: 0.00"

decimal total_expense "15, 2, default: 0.00"

decimal net_income "15, 2, default: 0.00"

uint64 closed_by FK "nullable, unsigned"

timestamp closed_at "default: now()"

text notes "nullable"

}

units ||--o{ chart_of_accounts : "unit_id"

chart_of_accounts ||--o{ chart_of_accounts : "parent_id"

units ||--o{ journal_entries : "unit_id"

```

```
users ||--o{ journal_entries : "posted_by"

journal_entries ||--o| journal_entries : "reversed_by_entry_id"

journal_entries ||--|{ journal_lines : "journal_id (cascade)"

chart_of_accounts ||--o{ journal_lines : "account_id"
```

C. Simpan Pinjam (Savings & Loans) Sub-System

Diagram siklus produk simpanan, jenis simpanan, pengajuan pinjaman, kontrak pinjaman aktif, angsuran jatuh tempo, dan pelunasan angsuran.

```
erDiagram
```

```
saving_types {
```

```
uint64 id PK "unsigned auto_increment"
```

```
varchar_20 code "unique"
```

```
varchar_100 name
```

```
boolean is_mandatory "default: false"
```

```
decimal min_amount "15, 2, default: 0.00"
```

```
decimal monthly_amount "15, 2, default: 0.00"
```

```
boolean is_withdrawable "default: true"
```

```
text description "nullable"
```

```
boolean is_active "default: true"
```

```
timestamp created_at "nullable"
```

```
timestamp updated_at "nullable"
```

```
}
```

```
savings {
```

```
uint64 id PK "unsigned auto_increment"
```

```
uint64 member_id FK "unsigned"
```

```
uint64 saving_type_id FK "unsigned"
```

```
decimal balance "15, 2, default: 0.00"
```

```
decimal total_deposit "15, 2, default: 0.00"
```

```
decimal total_withdraw "15, 2, default: 0.00"
```

```
timestamp created_at "nullable"
```

```
timestamp updated_at "nullable"
```

```
}
```

```
saving_transactions {
```

```
uint64 id PK "unsigned auto_increment"
```

```
uint64 savings_id FK "unsigned"
```

```
uint64 member_id FK "unsigned"
```

```

enum_saving_trx_type type

decimal amount "15, 2"

decimal balance_before "15, 2"

decimal balance_after "15, 2"

varchar_60 reference_no "unique"

text note "nullable"

uint64 processed_by FK "nullable, unsigned"

timestamp transaction_at

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

loan_products {

uint64 id PK "unsigned auto_increment"

varchar_20 code "unique"

varchar_100 name

decimal interest_rate "5, 2"

enum_loan_interest_method interest_method "default: flat"

int16 max_tenor

decimal max_amount "15, 2"

decimal min_amount "15, 2, default: 500000.00"

decimal admin_fee_pct "5, 2, default: 0.00"

decimal penalty_pct "5, 2, default: 0.00"

boolean requires_guarantor "default: false"

text requirements "nullable"

boolean is_active "default: true"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

```

```

loan_applications {

uint64 id PK "unsigned auto_increment"

uint64 member_id FK "unsigned"

uint64 loan_product_id FK "unsigned"

varchar_60 application_no "unique"

decimal amount_requested "15, 2"

int16 tenor_months

enum_loan_repayment_method repayment_method "default: cash"

text purpose

enum_loan_application_status status "default: draft"

timestamp submitted_at "nullable"

uint64 reviewed_by FK "nullable, unsigned"

timestamp reviewed_at "nullable"

uint64 approved_by FK "nullable, unsigned"

timestamp approved_at "nullable"

text rejection_note "nullable"

varchar_150 guarantor_name "nullable"

varchar_20 guarantor_phone "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

loans {

uint64 id PK "unsigned auto_increment"

uint64 application_id FK "unique, unsigned"

uint64 member_id FK "unsigned"

varchar_60 loan_no "unique"

decimal principal "15, 2"

decimal interest_rate "5, 2"

```



```

enum_loans_interest_method interest_method

decimal admin_fee "15, 2, default: 0.00"

int16 tenor_months

date disbursed_at

date first_due_date

date last_due_date

decimal monthly_installment "15, 2"

decimal outstanding_principal "15, 2"

decimal total_paid "15, 2, default: 0.00"

enum_loans_repayment_method repayment_method "default: cash"

enum_loans_status status "default: active"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

loan_schedules {

uint64 id PK "unsigned auto_increment"

uint64 loan_id FK "unsigned"

int16 installment_no

date due_date

decimal principal_due "15, 2"

decimal interest_due "15, 2"

decimal total_due "15, 2"

decimal principal_paid "15, 2, default: 0.00"

decimal interest_paid "15, 2, default: 0.00"

decimal penalty_paid "15, 2, default: 0.00"

timestamp paid_at "nullable"

enum_loan_schedules_status status "default: pending"

timestamp created_at "nullable"

```

```

timestamp updated_at "nullable"

}

loan_payments {

uint64 id PK "unsigned auto_increment"

uint64 loan_id FK "unsigned"

uint64 schedule_id FK "nullable, unsigned"

varchar_60 payment_no "unique"

decimal amount_paid "15, 2"

decimal principal_portion "15, 2"

decimal interest_portion "15, 2"

decimal penalty_amount "15, 2, default: 0.00"

enum_loan_payments_method payment_method

varchar_100 reference "nullable"

uint64 processed_by FK "nullable, unsigned"

timestamp paid_at

text note "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

members ||--o{ savings : "member_id"

saving_types ||--o{ savings : "saving_type_id"

savings ||--o{ saving_transactions : "savings_id"

members ||--o{ saving_transactions : "member_id"

users ||--o{ saving_transactions : "processed_by"

members ||--o{ loan_applications : "member_id"

loan_products ||--o{ loan_applications : "loan_product_id"

users ||--o{ loan_applications : "reviewed_by / approved_by"

```

```
loan_applications ||--o| loans : "application_id"

members ||--o{ loans : "member_id"

loans ||--|{ loan_schedules : "loan_id (cascade)"

loans ||--o{ loan_payments : "loan_id"

loan_schedules ||--o{ loan_payments : "schedule_id"

users ||--o{ loan_payments : "processed_by"
```

D. POS Retail & Inventory Sub-System

Diagram pengelolaan katalog produk, stok balance multi-lokasi retail/gudang, mutasi (kartu) stok, pemrosesan order penjualan kasir, transfer stok gudang, dan stok opname berkala.

```
erDiagram

products {

    uint64 id PK "unsigned auto_increment"

    uint64 unit_id FK "unsigned"

    uint64 category_id FK "unsigned"

    varchar_50 sku "unique"

    varchar_200 name

    text description "nullable"

    decimal purchase_price "15, 2, default: 0.00"

    decimal price "15, 2"

    decimal member_price "15, 2, nullable"

    int stock "default: 0"

    int min_stock "default: 0"

    varchar_20 unit_measure "default: pcs"

    varchar_255 image_path "nullable"

    boolean is_active "default: true"

    boolean is_online "default: true"

    boolean restock_requested "default: false"

    timestamp created_at "nullable"

    timestamp updated_at "nullable"

    timestamp deleted_at "nullable"

}

product_categories {

    uint64 id PK "unsigned auto_increment"

    uint64 parent_id FK "nullable, unsigned"

    varchar_100 name

    varchar_120 slug "unique"

    varchar_255 icon_url "nullable"

}
```

```

boolean is_active "default: true"

int sort_order "default: 0"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

orders {

uint64 id PK "unsigned auto_increment"

varchar_60 order_no "unique"

uint64 member_id FK "nullable, unsigned"

uint64 unit_id FK "unsigned"

enum_orders_channel channel "default: pos"

decimal subtotal "15, 2"

decimal discount "15, 2, default: 0.00"

decimal grand_total "15, 2"

enum_orders_payment_method payment_method "default: cash"

enum_orders_payment_status payment_status "default: unpaid"

enum_orders_order_status order_status "default: pending"

uint64 cashier_id FK "nullable, unsigned"

text note "nullable"

varchar_255 delivery_address "nullable"

timestamp ordered_at

timestamp paid_at "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

order_items {

uint64 id PK "unsigned auto_increment"

uint64 order_id FK "unsigned"

```

```

uint64 product_id FK "unsigned"

varchar_200 product_name

int qty

decimal unit_price "15, 2"

decimal discount "15, 2, default: 0.00"

decimal subtotal "15, 2"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

warehouse_locations {

uint64 id PK "unsigned auto_increment"

uint64 unit_id FK "unsigned"

varchar_50 location_code "unique"

varchar_150 location_name

enum_warehouse_location_type location_type "default: main"

text address "nullable"

boolean is_active "default: true"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

stock_balances {

uint64 id PK "unsigned auto_increment"

uint64 product_id FK "unsigned"

uint64 location_id FK "unsigned"

int qty_on_hand "default: 0"

int qty_reserved "default: 0"

int qty_available "default: 0"

timestamp updated_at "nullable"

```

```
}

stock_reorder_points {

uint64 id PK "unsigned auto_increment"

uint64 product_id FK "unique, unsigned"

int reorder_qty

int reorder_point

int lead_time_days "default: 7"

boolean is_active "default: true"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

stock_movements {

uint64 id PK "unsigned auto_increment"

uint64 product_id FK "unsigned"

enum_stock_movements_type type

int qty

int stock_before

int stock_after

varchar_100 reference "nullable"

text note "nullable"

uint64 created_by FK "nullable, unsigned"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

stock_transfer_orders {

uint64 id PK "unsigned auto_increment"

varchar_60 transfer_no "unique"

uint64 from_location_id FK "unsigned"
```

```

uint64 to_location_id FK "unsigned"

enum_stock_transfer_status status "default: pending"

uint64 requested_by FK "unsigned"

uint64 approved_by FK "nullable, unsigned"

uint64 transferred_by FK "nullable, unsigned"

timestamp approved_at "nullable"

timestamp transferred_at "nullable"

text notes "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"
}

stock_transfer_items {

uint64 id PK "unsigned auto_increment"

uint64 transfer_order_id FK "unsigned"

uint64 product_id FK "unsigned"

int qty_requested

int qty_transferred "default: 0"

timestamp created_at "nullable"
}

stock_opname {

uint64 id PK "unsigned auto_increment"

varchar_60 opname_no "unique"

uint64 location_id FK "nullable, unsigned"

date opname_date

enum_stock_opname_status status "default: draft"

text notes "nullable"

uint64 conducted_by FK "unsigned"

uint64 approved_by FK "nullable, unsigned"

```



```

timestamp approved_at "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

stock_opname_details {

uint64 id PK "unsigned auto_increment"

uint64 opname_id FK "unsigned"

uint64 product_id FK "unsigned"

int qty_system

int qty_physical

int variance

text notes "nullable"

timestamp created_at "nullable"

}

product_costing {

uint64 id PK "unsigned auto_increment"

uint64 product_id FK "unique, unsigned"

enum_product_costing_method costing_method "default: average_cost"

decimal current_cost "15, 2"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

units ||--o{ products : "unit_id"

product_categories ||--o{ products : "category_id"

product_categories ||--o{ product_categories : "parent_id"

units ||--o{ warehouse_locations : "unit_id"

products ||--o{ stock_balances : "product_id (cascade)"

```

```

warehouse_locations ||--o{ stock_balances : "location_id (cascade)"

products ||--o{ stock_reorder_points : "product_id (cascade)"

products ||--o{ stock_movements : "product_id"

users ||--o{ stock_movements : "created_by"

warehouse_locations ||--o{ stock_transfer_orders : "from_location / to_location"

stock_transfer_orders ||--|{ stock_transfer_items : "transfer_order_id (cascade)"

products ||--o{ stock_transfer_items : "product_id"

stock_opname ||--|{ stock_opname_details : "opname_id (cascade)"

products ||--o{ stock_opname_details : "product_id"

products ||--o{ product_costing : "product_id (cascade)"

orders ||--|{ order_items : "order_id (cascade)"

products ||--o{ order_items : "product_id"

members ||--o{ orders : "member_id"

units ||--o{ orders : "unit_id"

users ||--o{ orders : "cashier_id"

```

E. Procurement & Consignment (Titip Jual) Sub-System

Diagram pengelolaan logistik pembelian PO ke supplier, bukti penerimaan barang PO (good receipt), serta alur bagi hasil penjualan barang titipan (konsinyasi) milik supplier.

```
erDiagram

suppliers {

    uint64 id PK "unsigned auto_increment"

    varchar_50 supplier_code "unique"

    varchar_200 supplier_name

    varchar_150 contact_person "nullable"

    varchar_20 phone "nullable"

    varchar_100 email "nullable"

    text address "nullable"

    varchar_50 city "nullable"

    int16 payment_terms "nullable"

    int16 avg_delivery_days "nullable"

    boolean is_active "default: true"

    text notes "nullable"

    timestamp created_at "nullable"

    timestamp updated_at "nullable"

}

consignment_items {

    uint64 id PK "unsigned auto_increment"

    uint64 product_id FK "unsigned"

    uint64 supplier_id FK "unsigned"

    date consignment_date

    int qty_received

    int qty_sold "default: 0"

    int qty_returned "default: 0"

    varchar_500 return_reason "nullable"

    timestamp return_date "nullable"

    enum_consignment_status status "default: active"
```

```

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

consignment_payables {

uint64 id PK "unsigned auto_increment"

uint64 consignment_id FK "unsigned"

uint64 supplier_id FK "unsigned"

int qty_sold

decimal unit_price "15, 2"

decimal total_amount "15, 2"

enum_consignment_payable_status status "default: pending"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

consignment_settlements {

uint64 id PK "unsigned auto_increment"

uint64 payable_id FK "unsigned"

varchar_60 settlement_no "unique"

date settlement_date

decimal amount_paid "15, 2"

enum_payment_method_type payment_method

varchar_100 reference_no "nullable"

uint64 processed_by FK "unsigned"

timestamp created_at "nullable"

}

purchase_orders {

uint64 id PK "unsigned auto_increment"

uint64 supplier_id FK "unsigned"

```

```

varchar_60 po_no "unique"

date po_date

date expected_delivery

enum_po_status status "default: draft"

decimal subtotal "15, 2"

decimal tax_amount "15, 2, default: 0.00"

decimal total_amount "15, 2"

text notes "nullable"

uint64 created_by FK "unsigned"

uint64 approved_by FK "nullable, unsigned"

timestamp approved_at "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

purchase_order_items {

uint64 id PK "unsigned auto_increment"

uint64 po_id FK "unsigned"

uint64 product_id FK "unsigned"

int qty_ordered

int qty_received "default: 0"

decimal unit_price "15, 2"

decimal line_total "15, 2"

timestamp created_at "nullable"

}

good_receipts {

uint64 id PK "unsigned auto_increment"

uint64 po_id FK "unsigned"

uint64 supplier_id FK "unsigned"

```

```

varchar_60 gr_no "unique"

date gr_date

enum_gr_status status "default: received"

uint64 received_by FK "unsigned"

text notes "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

good_receipt_items {

uint64 id PK "unsigned auto_increment"

uint64 gr_id FK "unsigned"

uint64 product_id FK "unsigned"

int qty_received

int qty_accepted "default: 0"

int qty_rejected "default: 0"

text notes "nullable"

timestamp created_at "nullable"

}

suppliers ||--o{ consignment_items : "supplier_id"

products ||--o{ consignment_items : "product_id"

consignment_items ||--|{ consignment_payables : "consignment_id (cascade)"

suppliers ||--o{ consignment_payables : "supplier_id"

consignment_payables ||--|{ consignment_settlements : "payable_id (cascade)"

users ||--o{ consignment_settlements : "processed_by"

suppliers ||--o{ purchase_orders : "supplier_id"

purchase_orders ||--|{ purchase_order_items : "po_id (cascade)"

products ||--o{ purchase_order_items : "product_id"

```

```
purchase_orders ||--o{ good_receipts : "po_id"

suppliers ||--o{ good_receipts : "supplier_id"

good_receipts ||--|{ good_receipt_items : "gr_id (cascade)"

products ||--o{ good_receipt_items : "product_id"
```

F. Finance, Loyalty System, & SHU Sub-System

Diagram pengelolaan invoice hutang/piutang usaha non-pinjaman, perhitungan PPN/PPh penjualan, rekonsiliasi RAT tahunan, loyalitas poin belanja anggota, serta distribusi alokasi SHU.

```
erDiagram
```

```
accounts_payable {
```

```
uint64 id PK "unsigned auto_increment"
```

```
uint64 supplier_id FK "unsigned"
```

```
varchar_60 invoice_no "unique"
```

```
date invoice_date
```

```
date due_date
```

```
decimal subtotal "15, 2"
```

```
decimal tax_amount "15, 2, default: 0.00"
```

```
decimal total_amount "15, 2"
```

```
decimal amount_paid "15, 2, default: 0.00"
```

```
decimal amount_due "15, 2"
```

```
enum_ap_ar_status status "default: open"
```

```
text notes "nullable"
```

```
timestamp created_at "nullable"
```

```
timestamp updated_at "nullable"
```

```
}
```

```
accounts_payable_details {
```

```
uint64 id PK "unsigned auto_increment"
```

```
uint64 ap_id FK "unsigned"
```

```
uint64 product_id FK "nullable, unsigned"
```

```
varchar_255 description
```

```
int qty "default: 1"
```

```
decimal unit_price "15, 2"
```

```
decimal line_total "15, 2"
```

```
timestamp created_at "nullable"
```

```
}
```

```
accounts_receivable {
```



```

uint64 id PK "unsigned auto_increment"

uint64 member_id FK "nullable, unsigned"

varchar_150 customer_name

varchar_60 invoice_no "unique"

date invoice_date

date due_date

decimal subtotal "15, 2"

decimal tax_amount "15, 2, default: 0.00"

decimal total_amount "15, 2"

decimal amount_paid "15, 2, default: 0.00"

decimal amount_due "15, 2"

enum_ap_ar_status status "default: open"

decimal credit_limit "15, 2, nullable"

text notes "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

accounts_receivable_details {

uint64 id PK "unsigned auto_increment"

uint64 ar_id FK "unsigned"

uint64 product_id FK "nullable, unsigned"

varchar_255 description

int qty "default: 1"

decimal unit_price "15, 2"

decimal line_total "15, 2"

timestamp created_at "nullable"

}

tax_calculations {

```

```

uint64 id PK "unsigned auto_increment"

uint64 order_id FK "nullable, unsigned"

varchar_60 invoice_id "nullable"

enum_tax_type tax_type

decimal tax_percentage "5, 2"

decimal taxable_amount "15, 2"

decimal tax_amount "15, 2"

timestamp created_at "nullable"

}

shu_periods {

uint64 id PK "unsigned auto_increment"

uint64 unit_id FK "unsigned"

int period_year "year"

decimal total_revenue "15, 2, default: 0.00"

decimal total_expense "15, 2, default: 0.00"

decimal total_shu "15, 2, default: 0.00"

decimal shu_for_member "15, 2, default: 0.00"

decimal shu_for_reserve "15, 2, default: 0.00"

decimal shu_for_pengurus "15, 2, default: 0.00"

decimal shu_for_education "15, 2, default: 0.00"

enum_shu_periods_status status "default: draft"

timestamp calculated_at "nullable"

timestamp distributed_at "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

shu_distributions {

uint64 id PK "unsigned auto_increment"

```

```

uint64 shu_period_id FK "unsigned"

uint64 member_id FK "unsigned"

decimal savings_weight "10, 4, default: 0.0000"

decimal activity_weight "10, 4, default: 0.0000"

decimal shu_amount "15, 2"

enum_shu_distributions_status status "default: pending"

enum_shu_dist_disbursement_method disbursement_method "nullable"

timestamp disbursed_at "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"
}

rat_attendances {

uint64 id PK "unsigned auto_increment"

uint64 member_id FK "unsigned"

int year "year"

boolean is_present "default: true"

boolean voted "default: false"

timestamp attended_at "default: now()"

timestamp created_at "nullable"

timestamp updated_at "nullable"
}

loyalty_programs {

uint64 id PK "unsigned auto_increment"

varchar_50 program_code "unique"

varchar_150 program_name

text description "nullable"

decimal points_per_rupiah "10, 6"

decimal minimum_purchase "15, 2"

```

```

boolean is_active "default: true"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

loyalty_memberships {

uint64 id PK "unsigned auto_increment"

uint64 member_id FK "unsigned"

uint64 program_id FK "unsigned"

enum_membership_level membership_level "default: bronze"

decimal total_points "15, 2, default: 0.00"

decimal points_used "15, 2, default: 0.00"

decimal points_available "15, 2, default: 0.00"

date member_since

timestamp last_activity "nullable"

timestamp created_at "nullable"

timestamp updated_at "nullable"

}

loyalty_redemptions {

uint64 id PK "unsigned auto_increment"

uint64 membership_id FK "unsigned"

uint64 program_id FK "unsigned"

uint64 order_id FK "nullable, unsigned"

decimal points_redeemed "15, 2"

decimal discount_amount "15, 2"

timestamp redemption_date

timestamp created_at "nullable"

}

```

```

suppliers ||--o{ accounts_payable : "supplier_id"

accounts_payable ||--|{ accounts_payable_details : "ap_id (cascade)"

products ||--o{ accounts_payable_details : "product_id"

members ||--o{ accounts_receivable : "member_id"

accounts_receivable ||--|{ accounts_receivable_details : "ar_id (cascade)"

products ||--o{ accounts_receivable_details : "product_id"

units ||--o{ shu_periods : "unit_id"

shu_periods ||--|{ shu_distributions : "shu_period_id"

members ||--o{ shu_distributions : "member_id"

members ||--o{ rat_attendances : "member_id (cascade)"

members ||--o{ loyalty_memberships : "member_id (cascade)"

loyalty_programs ||--o{ loyalty_memberships : "program_id"

loyalty_programs ||--o{ loyalty_redemptions : "program_id"

```

G. Alur Korelasi Mutasi Keuangan & Akuntansi (Double-Entry Engine)

4. INTEGRITAS DATA & EFISIENSI QUERY

Untuk menjamin keandalan sistem berskala besar dengan transaksi tinggi, arsitektur database ini dilengkapi teknik optimasi performa dan validasi integritas:

A. Strategi Menjamin Integritas Data (Data Integrity)

1. Pola Saldo Sekuensial (Audit Trail):

Pada tabel **saving_transactions** dan **stock_movements**, kolom **balance_before** / **balance_after** dan **stock_before** / **stock_after** wajib diisi. Data ini dikunci saat transaksi berhasil dan tidak boleh diubah (*immutable*) guna mencegah manipulasi saldo historis.

2. Transaksi Database Terisolasi (Database Transaction):

Proses krusial seperti **Pembagian SHU Massal** dan **Persetujuan Stock Opname** dijalankan di dalam blok transaksi terisolasi (**prisma.\$transaction** di Prisma / **START TRANSACTION** di MySQL). Jika salah satu operasi gagal (misal: saldo simpanan gagal dikreditkan atau stok minus), seluruh perubahan dibatalkan (*Rollback*) sehingga tidak terjadi anomali data setengah jadi.

3. Cascading Foreign Keys Terseleksi:

- **ON DELETE CASCADE** diizinkan pada relasi detail anak seperti **order_items** ke **orders** dan **loan_schedules** ke **loans**. Jika master order/kontrak dihapus (atau dibatalkan sistem), detail item di bawahnya terhapus bersih.
- **ON UPDATE NoAction** atau **RESTRICT** diterapkan pada relasi master seperti **members** ke **units** atau **loans** ke **members** untuk mencegah hilangnya riwayat historis anggota yang memiliki sisa hutang atau saldo simpanan aktif.

B. Strategi Efisiensi Kueri (Query Efficiency & Indexing)

Aplikasi koperasi sering memuat laporan agregat per tanggal atau per anggota. Indeks database berikut dibuat secara khusus:

1. Indeks Komposit Penjualan & Anggota:

- **@@index([member_id, ordered_at])** pada tabel **orders**. Mempercepat kalkulasi partisipasi belanja tahunan anggota untuk SHU tanpa melakukan *full-table scan*.

2. Indeks Status & Jatuh Tempo Pembayaran:

- **@@index([due_date, status])** pada tabel **loan_schedules** dan **accounts_payable**. Sangat berguna bagi kueri scheduler harian untuk mencari cicilan jatuh tempo/terlambat (*overdue*) dan pembuatan notifikasi tagihan otomatis.

3. Kombinasi Unik Ganda (Composite Unique Constraints):

Mencegah inkonsistensi logis pada tingkat mesin database:

- **@@unique([unit_id, code])** pada **chart_of_accounts**: Mencegah duplikasi COA per unit.
- **@@unique([member_id, saving_type_id])** pada **savings**: Mengunci 1 anggota hanya mempunyai 1 rekening untuk tiap tipe tabungan.
- **@@unique([member_id, year])** pada **rat_attendances**: Menjamin anggota hanya melakukan 1 kali voting kehadiran RAT per tahun buku.